# LEVEL UP ACADEMY

Class 10 | GSEB | Science

**ANSWER KEY & MARKING SCHEME**

## Ch 11 Test (GSEB pattern): Electricity

*Chapter / Topic: Electricity*

*For teacher use only. Do not distribute to students.*

### SECTION A

**Q1.** A wire of resistance R is cut into five equal parts, which are then connected in parallel. If the equivalent resistance is R′, the ratio R/R′ is: **[1]**

**Answer:** (d) 25. Each part is R/5, and five in parallel give R′ = R/25.

**Q2.** The SI unit of potential difference is the: **[1]**

**Answer:** (a) volt

**Q3.** Which of the following does NOT represent electric power in a circuit? **[1]**

**Answer:** (b) $IR^2$

**Q4.** The length of a wire is doubled and its area of cross-section is halved. Its resistivity: **[1]**

**Answer:** (d) stays the same (resistivity depends only on the material and temperature)

**Q5.** The SI unit of electric charge is the: **[1]**

**Answer:** (a) coulomb

**Q6.** Two 4 Ω resistors are connected first in series and then in parallel. The ratio of the series resistance to the parallel resistance is: **[1]**

**Answer:** (c) 4. Series 8 Ω, parallel 2 Ω.

**Q7.** How much current does an 1100 W appliance draw from a 220 V supply? **[1]**

**Answer:** (c) 5 A. I = P/V = 1100/220

**Q8.** How much electrical energy does a 100 W bulb use in 10 hours? **[1]**

**Answer:** (b) 1 kWh. E = 0.1 kW × 10 h

**Q9.** An electric bulb is rated 220 V, 100 W. When it is operated on 110 V, the power consumed is: **[1]**

**Answer:** (a) 25 W. $R = V^2/P = 484\ \Omega$, and $P = 110^2/484 = 25$ W

### SECTION B

**Q10.** How much work is done in moving a charge of 2 C across two points that have a potential difference of 12 V? **[2]**

**Answer:** W = QV = 2 C × 12 V = 24 J

- Formula: 1
- 24 J: 1

**Q11.** A current of 0.5 A flows through a bulb for 10 minutes. How much charge flows through it? **[2]**

**Answer:** Q = It = 0.5 A × (10 × 60) s = 300 C

- Formula and time in seconds: 1
- 300 C: 1

**Q12.** Resistors of 4 Ω and 6 Ω are connected in series to a 12 V battery. Find the current in the circuit and the potential difference across the 4 Ω… **[2]**

**Answer:** R = 4 + 6 = 10 Ω. I = V/R = 12/10 = 1.2 A. V across 4 Ω = IR = 1.2 × 4 = 4.8 V.

- Current 1.2 A: 1
- 4.8 V: 1

### SECTION C

**Q13.** A copper wire has a diameter of 0.5 mm and a resistivity of $1.6 \times 10^{-8}$ Ω m. **[3]**

(i) $A = \pi d^2/4 = 3.14 \times (0.5 \times 10^{-3})^2/4 \approx 1.96 \times 10^{-7}$ m$^2$. $L = RA/\rho = (10 \times 1.96 \times 10^{-7})/(1.6 \times 10^{-8}) \approx 122.7$ m [2]

- Area: 1
- Length ≈ 122.7 m: 1

(ii) The area becomes 4 times, so the resistance becomes one-fourth (2.5 Ω). [1]

**Q14.** Resistors of 2 Ω, 3 Ω and 5 Ω are connected in series to a 10 V battery. **[3]**

(i) $R = 2 + 3 + 5 = 10$ Ω [1]

(ii) $I = V/R = 10/10 = 1$ A [1]

(iii) $V = IR = 1 \times 5 = 5$ V [1]

## SECTION D

**Q15.** In an experiment, a student measured the current (I) through a resistor for different potential differences (V). The readings were: I = 0.5, 1.0,… **[4]**

(i) V is directly proportional to I (Ohm's law). The V–I graph is a straight line through the origin. [1]

(ii) $R = V/I = 6.7/2.0 \approx 3.4$ Ω [1]

(iii) Resistance is proportional to length, so R doubles to about 6.7 Ω. $I = V/R = 6.7/6.7 \approx 1.0$ A [2]

- R doubles: 1
- I ≈ 1 A: 1